



Final assignment
Microsystems Design and Modelling (ET4260)
Part 2/3

Instructions. Document your calculations properly with derivations and screenshots when needed. Latex-generated PDF files are accepted.

Deadline for first submission of Part 2: 10th June 2026.

Deadline for final report submission: 24th June 2026.

Corrections made after the first submission are marked in [blue](#).

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FEM model of the phase shifter

This part models the phase shifter from Part 1 with finite elements in COMSOL 6.4. The device is the same except for one change: the fixed comb fingers on each end of the comb drive are now three times wider than the centre ones ($3W_c = 3\mu\text{m}$). Table 1 repeats the dimensions used here.

Table 1: Design parameters used in the FEM model.

Symbol	Value	Description
L	$100\mu\text{m}$	length of modulation section
g_0	200nm	modulation gap at rest
L_b	$75\mu\text{m}$	beam length
W_b	$5\mu\text{m}$	beam width
L_m	$40\mu\text{m}$	mass central section length
W_m	$30\mu\text{m}$	mass section width
h	$1\mu\text{m}$	comb-drive gap
d_0	$19\mu\text{m}$	comb overlap at rest
L_c	$20\mu\text{m}$	comb-finger length
W_c	$1\mu\text{m}$	comb-finger width
$3W_c$	$3\mu\text{m}$	bottom comb fingers on each end (new in Part 2)
t	200nm	thickness of suspended layer
N	40	number of comb-drive gaps
E_{Si}	170 GPa	Young's modulus of silicon
ρ_{Si}	2330kg/m^3	density of silicon
ν	0.28	Poisson ratio (assumed)
Q	83	quality factor imposed in Q3–Q6

The 2D model (top view) contains the suspended structure only: modulation bar ($L \times W_m$), central column, comb-spine mass plate, four suspension beams and the 20 moving + 21 fixed comb fingers. The waveguide is excluded, as instructed; only its 200 nm clearance above the bar enters the analysis (Q5). The outer beam ends carry fixed constraints. Silicon is isotropic with $E = 170\text{ GPa}$, $\rho = 2330\text{ kg/m}^3$ and an assumed $\nu = 0.28$ (the assignment specifies only E and ρ). Because the device layer is a 200 nm film with free top and bottom surfaces, plane stress is the appropriate 2D approximation; the layer thickness enters as the out-of-plane dimension. Figure 1 shows the mesh, refined around the $1\mu\text{m}$ comb features. All results below were checked on three refinement levels (4516, 12709 and 44563 elements).

One bookkeeping difference with Part 1 matters throughout this report. The lumped mass used there, $m = \rho t(LW_m + L_mW_m + NL_cW_c) = 2.33\text{ pg}$, counts one $100 \times 30\mu\text{m}$ plate and 40 fingers. The drawn geometry has two such plates (modulation bar *and* comb spine) and 20 moving fingers, which gives 3.54 pg before counting any beam mass. This factor of 1.5 in mass, together with the FEM spring constant, explains most of the differences with the Part 1 estimates in Q2, Q3 and Q5.

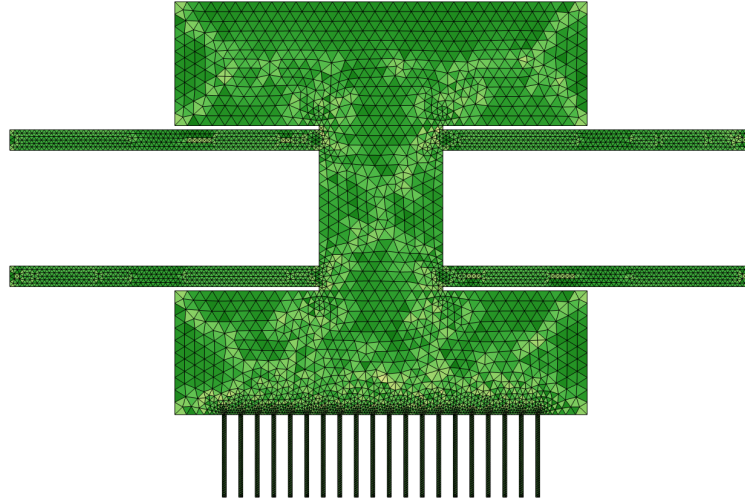


Figure 1: Mesh of the mechanical model (normal refinement, 12 709 elements). The element size follows the geometry: coarse on the $100 \times 30 \mu\text{m}$ plates, fine in the comb region where the features are $1 \mu\text{m}$.

1 Mechanical model and dynamics

Q1 – Spring constant from a stationary analysis

A $1 \mu\text{N}$ test force is applied to the mass plate along the motion axis and the spring constant follows from $k = F/u_y$. Table 2 lists the result for three mesh densities. The value decreases monotonically with refinement, as expected for displacement-based finite elements, and changes by only 0.2% over a factor ten in element count.

Table 2: Mesh convergence of the spring constant.

mesh	elements	k (N/m)
coarse	4 516	36.99
normal	12 709	36.87
fine	44 563	36.79

The FEM gives $k = 36.8 \text{ N/m}$, 8.7% below the Part 1 estimate of 40.30 N/m from four fixed-guided beams,

$$k_{\text{P1}} = \frac{4E_{Si} t W_b^3}{L_b^3}. \quad (1)$$

To find where the 8.7% comes from, the model was re-solved with the plates and column made artificially rigid ($E \times 1000$ everywhere except the beams). That returns $k = 39.83 \text{ N/m}$, within 1.1% of the analytical value. So the discrepancy splits into two parts: about 7.6% is compliance of the plates and column, which the analytical model treats as rigid (the guided beam end rotates slightly when the “rigid” body deforms), and the remaining 1.1% is shear deformation and 2D continuum effects that Euler-Bernoulli beam theory leaves out. With $W_b/L_b = 1/15$ the beams are slender, so a small shear contribution is consistent.

Q2 – Eigenfrequency analysis and mode shapes

Table 3 lists the first five eigenfrequencies; Fig. 2 shows the modal displacement maps. The fundamental mode at 496.3 kHz is the actuation mode: the shuttle translates along y and the four beams take the S-shape of the fixed-guided boundary condition. Mode 2 is an in-plane

rocking of the shuttle. Modes 3–5 sit within 4 kHz of each other and belong to the comb fingers: 20 nearly identical 20 μm cantilevers form a band of collective bending modes. A single-finger estimate, $f = 0.56 (W_c/L_c^2) \sqrt{E/\rho} \approx 3.4 \text{ MHz}$, lands on the band. Frequencies changed by less than 0.1% between the normal and fine mesh.

Table 3: First five eigenfrequencies (fine mesh).

mode	character	f (kHz)
1	shuttle translation along y (actuation mode)	496.26
2	in-plane rocking of the shuttle	2542.7
3	finger band, collective in-phase bending	3308.1
4	finger band, alternating pattern	3311.2
5	finger band, higher spatial pattern	3311.2

Part 1 predicted $f_0 = 661.9 \text{ kHz}$, 33% above the FEM value. The ratio is fully accounted for by the stiffness and mass corrections:

$$\frac{f_{\text{FEM}}}{f_{\text{P1}}} = \sqrt{\frac{k_{\text{FEM}}}{k_{\text{P1}}} \cdot \frac{m_{\text{P1}}}{m_{\text{eff}}}} = \sqrt{0.913 \times 0.613} = 0.748 \quad \Rightarrow \quad 495 \text{ kHz}. \quad (2)$$

Here $m_{\text{eff}} = k/(2\pi f_1)^2 = 3.79 \text{ pg}$: the figure-true shuttle mass of 3.54 pg plus about 36% of the beam mass, close to the textbook fraction for a distributed fixed-guided beam.

One limitation should be stated. A 2D in-plane model cannot show out-of-plane modes. The out-of-plane bending stiffness of the beams is $(t/W_b)^2 = 1/625$ of the in-plane one, so the physical device has an out-of-plane mode near 20 kHz, far below all modes in Table 3. The assignment asks for the 2D model, so this mode is outside the scope here, but a 3D design check would have to start with it.

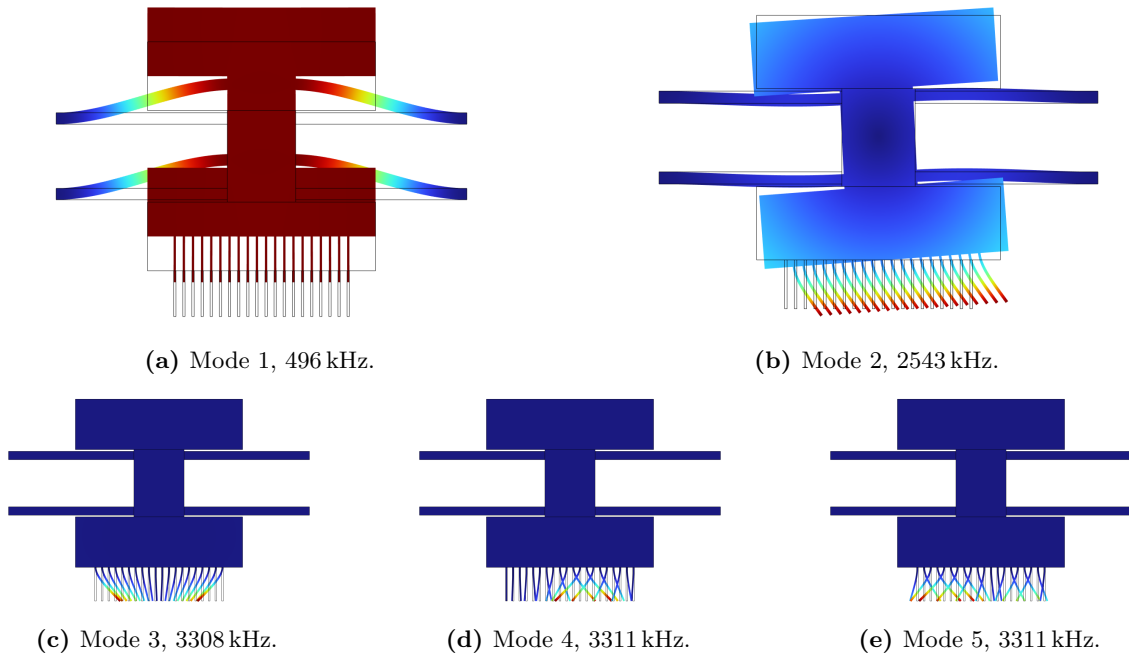


Figure 2: Modal displacement maps of the first five modes (colour: total displacement, normalised per mode). Mode 1 is the actuation mode; modes 3–5 are collective comb-finger modes.

Q3 – Rayleigh damping and force-step response

The assignment asks for $Q = 83$. Rayleigh damping gives a modal damping ratio

$$\zeta(f) = \frac{\alpha_{dM}}{4\pi f} + \beta_{dK} \pi f, \quad (3)$$

with two free parameters but only one target, $\zeta(f_1) = 1/(2Q)$. I chose pure stiffness damping, $\alpha_{dM} = 0$ and

$$\beta_{dK} = \frac{1}{2\pi f_1 Q} = \frac{1}{2\pi \cdot 496.26 \text{ kHz} \cdot 83} = 3.87 \times 10^{-9} \text{ s}, \quad (4)$$

because β -damping grows with frequency and therefore also suppresses the 3.3 MHz finger band; mass-proportional damping would have left those modes nearly undamped on top of the response.

The transient applies a 1 μN step to the mass at $t = 0$ and runs to 400 μs with a relative tolerance of 10^{-5} . Figure 3 shows the result. The response rings at f_1 , overshoots to $1.96\times$ the static value (the ideal underdamped limit is 2), and decays inside the analytic envelope $e^{-\zeta\omega_1 t}$. The final value of 27.11 nm matches $F/k = 27.12 \text{ nm}$ to 0.05%.

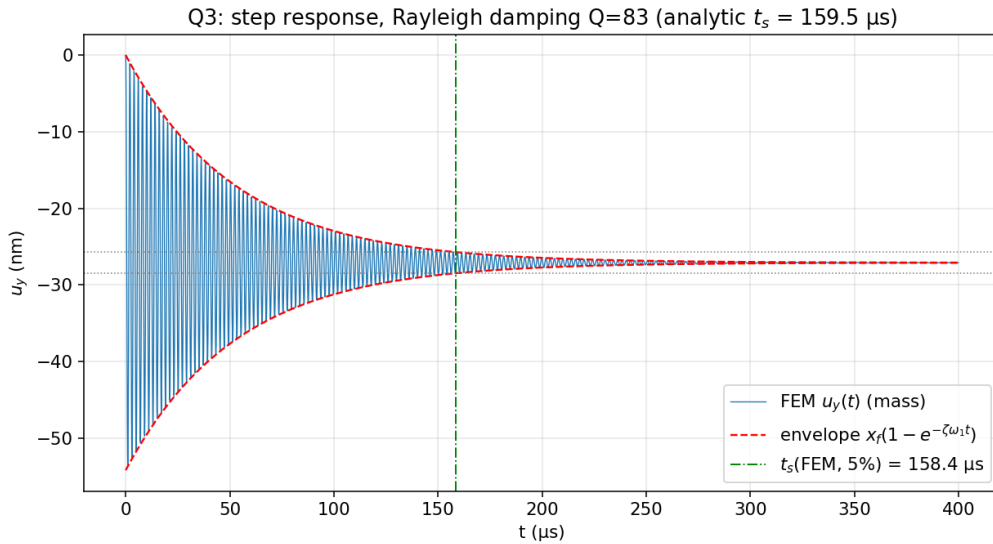


Figure 3: Step response to a 1 μN force with Rayleigh damping set for $Q = 83$. Red dashes: analytic envelope $x_f(1 \pm e^{-\zeta\omega_1 t})$; dotted lines: $\pm 5\%$ band; green line: measured 5% settling time.

The measured 5% settling time is 158.4 μs . The analytic value with the FEM frequency, $t_s = \ln(20)/(\zeta\omega_1) = 159.5 \mu\text{s}$, agrees to 0.7%, which also confirms that the integrator adds no significant numerical damping at this tolerance. Part 1 predicted 139 μs . The formula is the same; the inputs differ. Part 1 used its computed $Q = 96.8$ where the assignment now fixes $Q = 83$, and its ω_0 was 33% high (Q2). Since $t_s \propto Q/\omega$, the two corrections nearly cancel and the predictions end up closer than the inputs would suggest.

2 Electrostatic transduction

Following the tutorials referenced in the assignment, the electrostatics is added as follows. An air domain fills the cavity between the comb and the stator. The whole shuttle is a *domain terminal* at 0 V, which also returns its total charge and hence the capacitance. The stator floor and the fixed-finger walls carry the drive potential V_d . An *Electromechanical Forces* coupling applies the Maxwell stress to the solid, and a *Moving Mesh* node (hyperelastic smoothing) deforms the air mesh with the structure, with roller conditions at the two side openings. Figure 4 shows the

potential at $V_d = 50$ V: the drop concentrates in the 40 flank gaps and in the two $1\text{ }\mu\text{m}$ tip gaps of each finger.

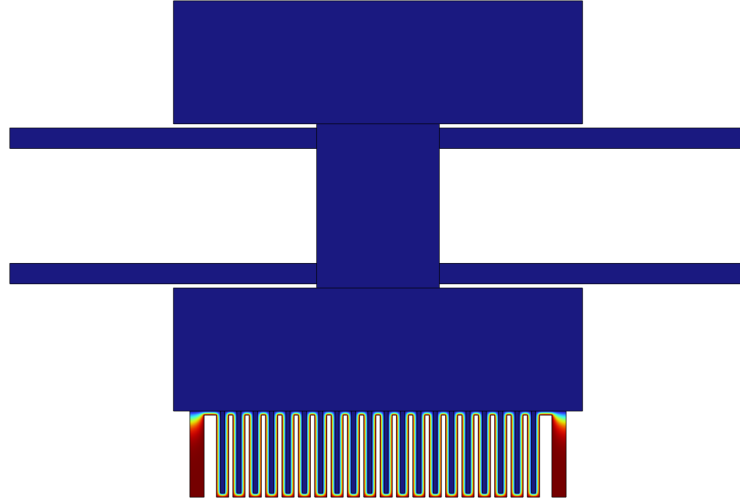


Figure 4: Electric potential at $V_d = 50$ V. Blue: shuttle terminal at 0 V; red: stator and fixed fingers at V_d . The wide end fingers ($3W_c$) are visible at both edges of the comb.

Q4a – Displacement versus voltage, 0–150 V

A stationary study sweeps V_d from 0 to 150 V in 10 V steps, each solution continuing from the previous one. Figure 5 plots the FEM displacement together with the Part 1 prediction $x = V^2 N \epsilon_0 t / (2kh)$.

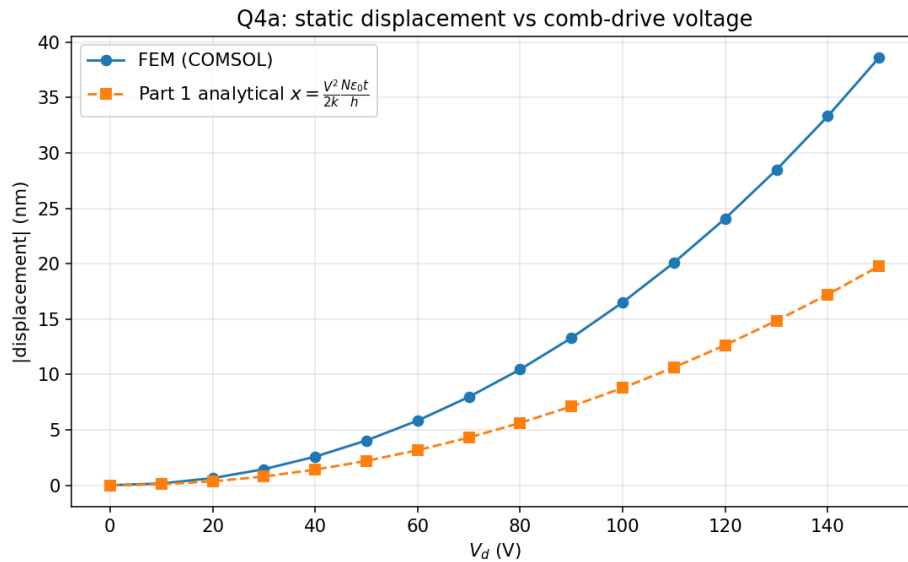


Figure 5: Static displacement of the mass versus comb-drive voltage: FEM (blue) and the Part 1 ideal-comb expression (orange). Both are quadratic in V ; the FEM force is larger by a nearly constant factor.

Both curves are quadratic, but the FEM displacement is $1.84\times$ the analytical one at 50 V and $1.95\times$ at 150 V. The factor splits cleanly in two:

- $\times 1.09$ from the softer FEM spring (Q1);

- $\times 1.68$ from the electrostatic force gradient. The ideal-comb formula counts only the flank capacitance, $dC/dx = N\epsilon_0 t/h = 7.08 \times 10^{-11} \text{ F/m}$. In this geometry the comb is engaged to $d_0 = 19$ of $20 \mu\text{m}$, so two families of $1 \mu\text{m}$ tip gaps remain: mover tips facing the stator floor, and fixed-finger tips facing the mass plate. Both close as the shuttle moves down and add roughly $\epsilon_0 W_c t/g^2$ per tip. The force-implied gradient from the FEM at 50 V is $1.19 \times 10^{-10} \text{ F/m}$, i.e. the tip gaps contribute almost as much force as all the flanks together.

The terminal capacitance supports the same picture: $C(0) = 1.51 \text{ fF}$ against the parallel-plate value $N\epsilon_0 d_0 t/h = 1.35 \text{ fF}$ (the 12% excess is fringing), and C rises with voltage as the comb engages deeper. The slow growth of the FEM/analytical ratio with voltage is the $1/g^2$ tip nonlinearity beginning to show – the same mechanism that ends in pull-in.

Q4b – Pull-in voltage

The voltage is raised in 5 V continuation steps until the stationary solver stops converging. Because each step starts from the previous equilibrium, the solution follows the physical stable branch, and the first voltage with no stable equilibrium is the pull-in point. The sweep converges up to 222.2 V (displacement 94 nm) and diverges beyond it:

$$V_{\text{PI}}^{\text{FEM}} = 222 \text{ V to } 227 \text{ V} \quad (5)$$

against 347 V from the Part 1 tip-gap model (Fig. 6).

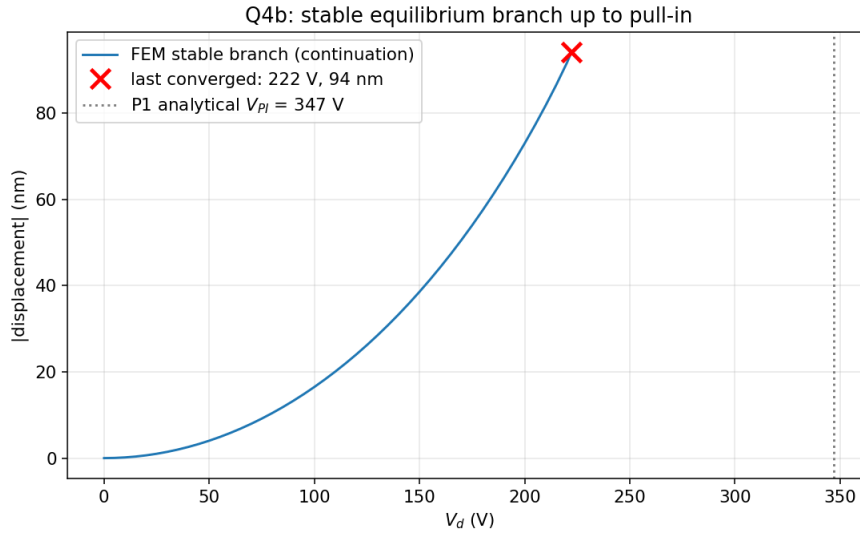


Figure 6: Stable equilibrium branch from the continuation sweep. The branch ends where no stable solution exists; the last converged point is 222.2 V at 94 nm .

Two things stand out: the FEM pull-in is 36% below the Part 1 value, and it happens at 94 nm , far short of the $g/3 \approx 333 \text{ nm}$ rule of thumb for a $1 \mu\text{m}$ gap. Correcting Part 1's formula with the FEM stiffness and force gradient, $V_{\text{PI}} \approx 347 \sqrt{(k_{\text{FEM}}/k_{\text{P1}}) [(dC/dx)_{\text{P1}}/(dC/dx)_{\text{FEM}}]} \approx 240 \text{ V}$, gets close to the bracket but still misses what actually happens.

To identify the mechanism, a prestressed eigenfrequency study was run at increasing bias: stationary solution first, then the eigenproblem linearised around it. An instability corresponds to an eigenfrequency reaching zero. Figure 7 shows the normalised stiffness $(f/f_0)^2$ of the three lowest modes. The axial actuation mode barely softens (it would reach zero near 445 V), but *mode 2 – the rocking mode – collapses*: 2545 kHz at rest, 509 kHz at 220 V , with $f_2^2 \rightarrow 0$ extrapolating to 224 V . That matches the observed divergence at 222 V .

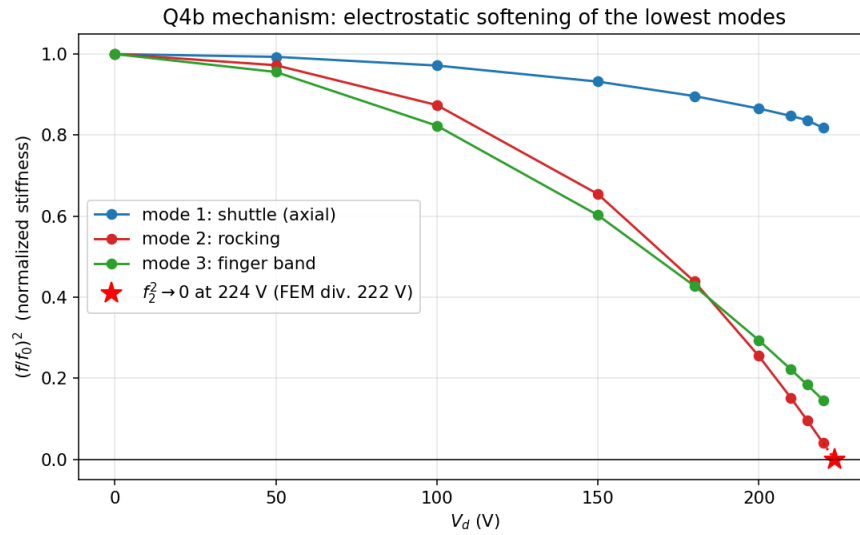


Figure 7: Electrostatic softening of the three lowest modes from the prestressed eigenfrequency study. The rocking mode (red) reaches zero stiffness at ≈ 224 V, the axial mode (blue) only at ≈ 445 V.

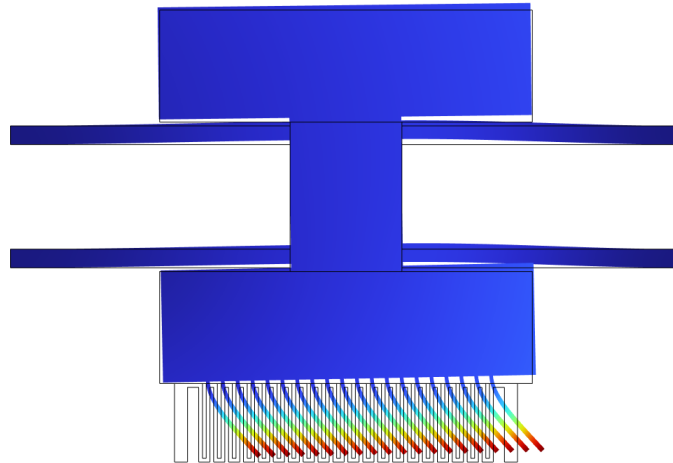


Figure 8: Shape of the critical mode at $V_d = 220$ V: a slight rotation of the shuttle with all comb fingers bending sideways in unison. This is the mode that goes unstable at pull-in.

The physics is side pull-in. Each finger sits between two $1\text{ }\mu\text{m}$ gaps, a laterally balanced but unstable arrangement whose negative stiffness grows as $V^2 d_0/h^3$. A small rotation of the shuttle moves every finger toward one gap wall at once, on long moment arms (Fig. 8). At 224 V this rotational softening overwhelms the suspension. The Part 1 model is a single-DOF axial model; it cannot represent this failure mode at all, which is why it overpredicts V_{PI} by 56% and also predicts the wrong kind of instability. A design fix would stiffen the rotation (larger beam separation) or the fingers themselves (shorter or wider).

A note on method: the assignment suggests the displacement-controlled continuation from the COMSOL pull-in tutorial (a global equation solving for the voltage that holds a prescribed displacement). I reproduced that setup – point probe in the material frame, baseline study with the global equation disabled, segregated solver groups – but COMSOL 6.4 fails to assemble freshly generated solver sequences for global equations that reference the displacement field (NaN in the moving-mesh residual). The same failure occurs on COMSOL’s own tutorial file when the study is recreated, while the shipped, older solver sequence runs fine. The voltage-continuation

method used above does not involve a global equation and is unaffected; its bracket is confirmed independently by the eigenfrequency extrapolation.

3 Acceleration limit and voltage-step response

Q5 – Maximum acceleration before mass-waveguide contact

An acceleration a along the motion axis loads every suspended part with a body force ρa per unit volume. Contact occurs when the modulation bar crosses the 200 nm gap to the waveguide, $u_y = +g_0$. For each bias the model is solved at two accelerations, $a_{\max}$ follows from linear scaling, and a third solve at $a_{\max}$ verifies the result; the verification residuals stay below 0.2%, so the linear treatment is justified. Table 4 and Fig. 9 give the results.

Table 4: Maximum tolerable DC acceleration toward the waveguide.

V_d	rest position u_y	$a_{\max}$ (m/s ²)	in g
0 V	0	1.903×10^6	193 950
1 V	−0.002 nm	1.903×10^6	193 950
10 V	−0.160 nm	1.904×10^6	194 060
50 V	−4.03 nm	1.930×10^6	196 690

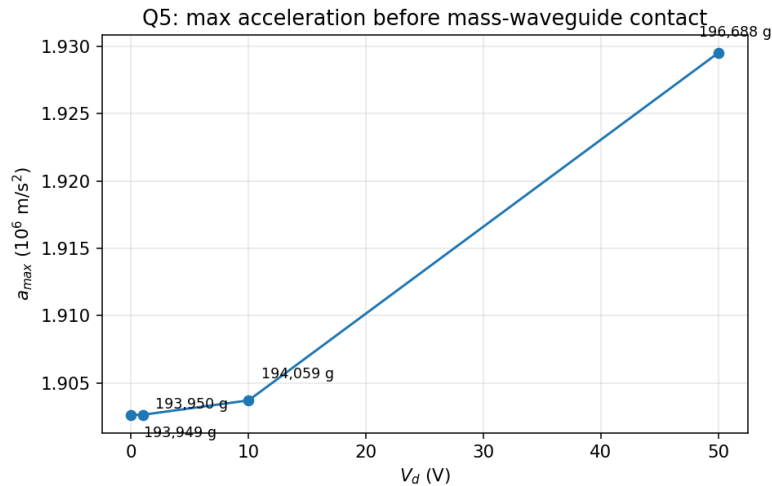


Figure 9: Maximum acceleration versus bias voltage. The bias moves the shuttle away from the waveguide, so the tolerance grows slightly with V_d .

The bias dependence is weak but has a definite sign: $a_{\max}$ *increases* with V_d , by 1.4% at 50 V. The comb pulls the shuttle away from the waveguide, which adds travel margin, and the tip-gap attraction weakens as the shuttle moves up, which helps further. Part 1 assumed the opposite sign (bias displacement toward the waveguide) and should be corrected on this point in the final report.

The magnitude also differs from Part 1's 3.46×10^6 m/s². Dividing kg_0 by the FEM limit gives an effective inertia of 3.88 pg: the figure-true shuttle plus about half the beam mass, since the distributed body load also bends the beams directly. Part 1's smaller mass (2.33 pg) and stiffer spring account for the rest. Either way, the static shock tolerance is around $2 \times 10^5 g$; as in Part 1, the practical concern is excitation near resonance, not static acceleration.

Q6 – Voltage-step response, 1 V and 50 V

The drive electrode receives a smoothed voltage step (rise time $1\ \mu\text{s}$, well below the $2\ \mu\text{s}$ oscillation period but gentle enough for the coupled solid–electrostatics–moving-mesh integrator, which fails on nanosecond-scale steps). Rayleigh damping is the same as in Q3. Figure 10 shows both responses; Fig. 11 overlays them with the Q3 force step, normalised to their final values.

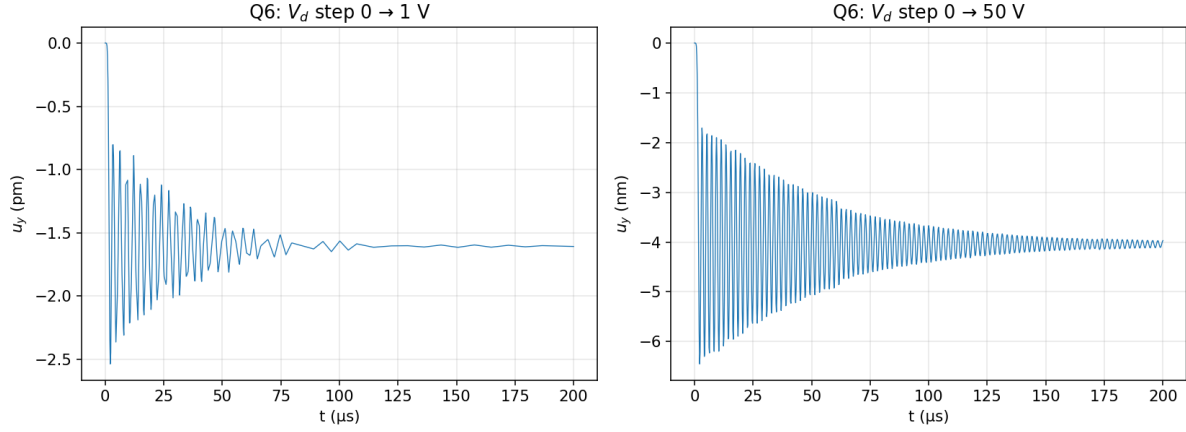


Figure 10: Step responses for V_d : $0 \rightarrow 1\ \text{V}$ (left, picometre scale) and $0 \rightarrow 50\ \text{V}$ (right, nanometre scale). Note the 2500 : 1 amplitude ratio for a 50 : 1 voltage ratio.

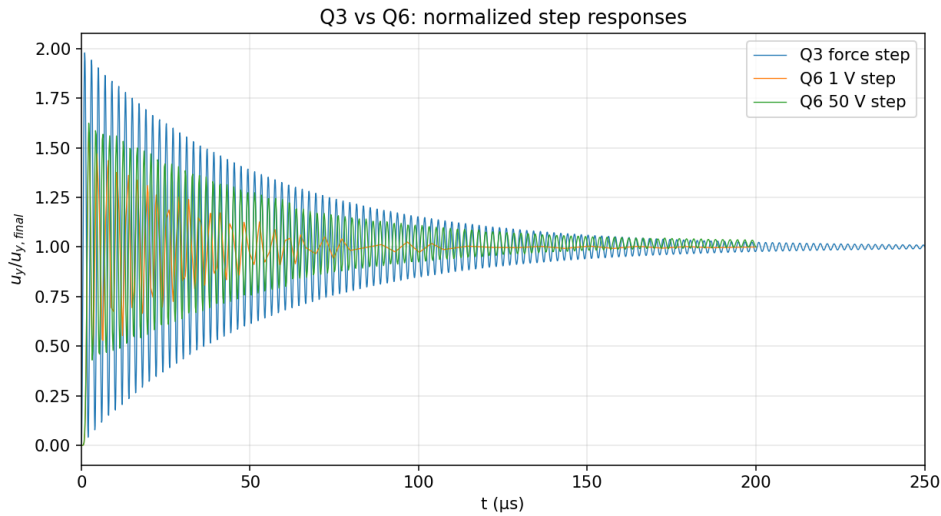


Figure 11: Normalised comparison of the Q3 force step and the Q6 voltage steps. All three ring at f_1 and settle on the same time scale; the voltage steps overshoot less because of the $1\ \mu\text{s}$ input rise.

The comparison with Q3 comes down to three observations.

First, the dynamics are identical. Both responses ring at $f_1 = 496\ \text{kHz}$ and the $50\ \text{V}$ step settles in about $150\ \mu\text{s}$, matching Q3's $158\ \mu\text{s}$ within the numerical accuracy of the coupled transient. At these voltages the device operates far below pull-in ($222\ \text{V}$), so the electromechanical system is effectively linear and a voltage step acts exactly like a force step of size $F = \frac{1}{2}V^2 dC/dx$. The bias softening measured in Q4b supports this: at $50\ \text{V}$ the mode-1 frequency shifts by only -0.3% , invisible in the ringing.

Second, the input scaling is quadratic, and that is the real difference between the two experiments. Final values are $-1.61\ \text{pm}$ at $1\ \text{V}$ and $-3.97\ \text{nm}$ at $50\ \text{V}$ – a factor 2466 for a factor 50 in voltage,

against 2500 expected from V^2 (the static values from Q4a are reproduced to within 1.7%). A force input scales linearly; a voltage input does not. For the phase-shifter application this is the linearity limit of the actuator, independent of any mechanical nonlinearity.

Third, the overshoot is $1.62\times$ instead of Q3's $1.96\times$. This is the input, not the plant: the $1\text{ }\mu\text{s}$ rise filters the step. The same filtering appears identically in both Q6 traces.

One numerical remark. The 50 V trace is quantitative (envelope decay within 7% of $\zeta\omega_1$). The 1 V response, at 1.6 pm, sits below the integrator's effective absolute error floor, and its apparent decay is dominated by numerical dissipation even at a relative tolerance of 10^{-5} – re-running with tighter settings did not change it. The waveform and final value are correct; the decay is not. Since the system is linear at these amplitudes, the physical 1 V transient is the 50 V trace scaled by $1/2500$, with the same settling time. Resolving picometre transients directly would need manual per-variable absolute tolerances, which is beyond what the comparison requires.